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CariSurg MedTech Pathways Healthcare AI Program
Week 2: Project Setup & Documentation
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Research Methodology Note

Papers were sourced using PubMed. ChatGPT was used to help extract the Q1–Q4 structure from each abstract, discussion and conclusion. After confirming each paper was relevant to the research focus, the full text was reviewed and summaries refined. The problem statement was designed around the research gaps identified in the previous work literature, particularly the limited validation of these systems in Caribbean emergency department settings. The reference lists are made by Zotero using the Vancouver citation style.

CariSurg MedTech Pathways Research Project

Preliminary Proposal

Project Title: AI-Assisted Early Sepsis Detection in the Emergency Department: A Pilot Study at Mercer General Hospital

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Statement of the Problem:

Sepsis is a severe reaction to infection responsible for approximately 11 million deaths worldwide each year [1]. Early recognition and treatment are critical, as studies indicate that each hour of delay in antibiotic treatment can increase mortality risk by 4–9% [2]. AI tools such as NAVOY® Sepsis have shown promise in detecting sepsis earlier in intensive care units in Northern Europe [3]. However, these tools have not been evaluated in Caribbean emergency departments, where triage nurses rely mainly on clinical judgment and delays in sepsis recognition remain a real and ongoing challenge.

Previous Work:

Several machine learning tools have been developed for early sepsis detection, but most have been tested in North American, European and Asian hospitals rather than Caribbean emergency departments.

Summary 1 - Sepsis Prediction at Emergency Department Triage Using Natural Language Processing: Retrospective Cohort Study

Brann et al. (2024) investigated whether a machine learning model could predict sepsis at the time of emergency department triage using nursing triage notes and patient data recorded on arrival. The model was trained and tested on over one million adult emergency department encounters collected from four academically affiliated US hospitals between 2015 and 2021. The system detected 36.1% of sepsis cases three hours before the first antibiotic treatment, 49.9% two hours before, and 68.3% one hour before, indicating strong overall predictive performance (AUC = 0.94). However, the authors note that the model

was developed exclusively in US academic hospitals with standardised nursing documentation, making its performance in settings such as Caribbean emergency departments uncertain.

Summary 2 - Early prediction of sepsis in intensive care patients using the machine learning algorithm NAVOY® Sepsis, a prospective randomized clinical validation study

Persson et al. (2024) investigated whether the machine learning algorithm NAVOY® Sepsis could predict sepsis in ICU patients before clinical recognition. The study evaluated 304 adult intensive care unit (ICU) patients at Skåne University Hospital in Sweden using vital signs, blood gas measurements, and laboratory data. Results showed that NAVOY® Sepsis could predict sepsis approximately three hours before clinical diagnosis with good performance (accuracy of 79%, sensitivity of 80%, and specificity of 78%). However, the authors note that the study was conducted in a single ICU and delays in manual data entry during COVID-19 affected the timing of alerts, limiting their usefulness for clinical decision-making.

Summary 3 - External Validation of a Widely Implemented Proprietary Sepsis Prediction Model in Hospitalized Patients

Wong et al. (2021) evaluated the performance of the Epic Sepsis Model, a machine learning tool widely used across US hospitals, by testing it on electronic health record data from 27,697 patients at Michigan Medicine, a different healthcare system from where the model was originally trained. The researchers assessed how accurately the model predicted sepsis, how many cases it detected, and how often it generated alerts using scores calculated every 15 minutes. Results showed poor performance, the model missed 67% of sepsis cases and generated alerts for 18% of all hospitalised patients, a false-positive rate so disruptive that clinical staff began ignoring the alerts altogether. The authors acknowledged that the study was limited to one hospital using one alert setting, so it is unclear how the model would perform in a different healthcare system, including a Caribbean emergency department like Mercer General Hospital.

Summary 4 - Global, regional, and national sepsis incidence and mortality, 1990–2017: analysis for the Global Burden of Disease Study

Rudd et al. (2020) estimated the global scale of sepsis using death records from 109 million individuals and 309 million hospital records across 195 countries. The study found that in 2017 alone, there were approximately 48.9 million sepsis cases worldwide and 11 million sepsis-related deaths, accounting for nearly one in five deaths globally. Although the overall rates of sepsis cases and deaths decreased between 1990 and 2017, the highest rates remained in lower-income regions including sub-Saharan Africa and south Asia. The authors noted that limited high-quality hospital and mortality data from many lower-income regions may lead to underestimation of the true global burden of sepsis.

Summary 5 - Machine Learning and Deep Learning Models for Early Sepsis Prediction: A Scoping Review

Shanmugam et al. (2025) reviewed 13 studies on machine and deep learning systems for early sepsis prediction. According to their research, many models detected sepsis earlier and more accurately than traditional scoring systems, although only one had been tested in a real hospital setting. The authors also

found that most systems relied on highly structured electronic health records and continuous digital monitoring, which may not be available in hospitals that still rely heavily on paper records.

Summary 6 - Development and validation of a screening tool for sepsis without laboratory results in the emergency department: a machine learning study

Jiang et al. (2025) developed a machine learning model called qSepsis to identify sepsis during emergency department triage using only information available before laboratory results, including vital signs, symptoms, triage level and mode of arrival. The model was trained on more than 620,000 emergency department records from a large hospital in China and later tested using a large US emergency department database (MIMIC-IV database). The researchers found that qSepsis predicted sepsis more accurately than three traditional clinical screening tools (SOFA, MEWS, and SIRS). The authors also noted that the system could be useful in emergency departments with limited laboratory access. However, both datasets came from large hospitals with advanced electronic health record systems, so the model's performance in smaller Caribbean hospitals with paper-based record systems remains unknown.

Summary 7 - Interpretable machine learning for predicting sepsis risk in emergency triage patients

Liu et al. (2025) tested whether adding demographic information, medical history, and chief complaints to routine vital signs improved sepsis prediction during emergency department triage. Using patient records from a large US hospital database (MIMIC-IV database), the researchers compared eight machine learning models and found that models using broader triage information predicted sepsis more accurately than models using vital signs alone. The study showed that combining multiple types of triage data improved early sepsis detection in emergency department patients. However, the authors note that the model was developed using data from a single US academic hospital database and only tested in a US hospital setting.

Proposed Solution

I propose a 12-week pilot to test whether a machine learning model trained on a Mercer General emergency department dataset can identify at least 50% of sepsis cases at least two hours before clinical diagnosis. The model will use only information typically available at triage, such as vital signs, age, and chief complaint, to reflect the conditions of a Caribbean emergency department where laboratory results are not always immediately available.

Gaps: Brann et al. (2024) demonstrated strong sepsis prediction at triage across four large US academic hospitals but did not test performance in settings where nursing documentation is brief and digital infrastructure is inconsistent, conditions common in many Caribbean emergency departments. Persson et al. (2024) validated NAVOY® Sepsis in a well-resourced Swedish ICU but stated that generalisability to lower-resource or non-European settings remains unproven, leaving its relevance to Caribbean emergency departments untested.

References:

1. Rudd KE, Johnson SC, Agesa KM, Shackelford KA, Tsoi D, Kievlan DR, et al. Global, regional, and national sepsis incidence and mortality, 1990–2017: analysis for the Global Burden of Disease Study. *The Lancet*. 2020 Jan 18;395(10219):200–11. doi:[10.1016/S0140-6736\(19\)32989-7](https://doi.org/10.1016/S0140-6736(19)32989-7) PubMed PMID: 31954465.
2. Balch B. Sepsis is the third leading cause of death in U.S. hospitals. But quick action can save lives. Association of American Medical Colleges (AAMC) News. 2023 Oct 10. <https://www.aamc.org/news/sepsis-third-leading-cause-death-us-hospitals-quick-action-can-save-lives>
3. Persson I, Macura A, Becedas D, Sjövall F. Early prediction of sepsis in intensive care patients using the machine learning algorithm NAVOY® Sepsis, a prospective randomized clinical validation study. *Journal of Critical Care*. 2024 Apr 1;80:154400. doi:[10.1016/j.jcrc.2023.154400](https://doi.org/10.1016/j.jcrc.2023.154400)
4. Brann F, Sterling NW, Frisch SO, Schrager JD. Sepsis Prediction at Emergency Department Triage Using Natural Language Processing: Retrospective Cohort Study. *JMIR AI*. 2024 Jan 25;3(1):e49784. doi:[10.2196/49784](https://doi.org/10.2196/49784)
5. Wong A, Otles E, Donnelly JP, Krumm A, McCullough J, DeTroyer-Cooley O, et al. External Validation of a Widely Implemented Proprietary Sepsis Prediction Model in Hospitalized Patients. *JAMA Intern Med*. 2021 Aug 1;181(8):1065–70. doi:[10.1001/jamainternmed.2021.2626](https://doi.org/10.1001/jamainternmed.2021.2626)
6. Shanmugam H, Rawat S, Airen L. Machine Learning and Deep Learning Models for Early Sepsis Prediction: A Scoping Review. *Indian Journal of Critical Care Medicine*. 2025 Jun 5;29(6):516–24. doi:[10.5005/jp-journals-10071-24986](https://doi.org/10.5005/jp-journals-10071-24986)
7. Jiang S, Dai S, Li Y, Zhou X, Jiang C, Tian C, et al. Development and validation of a screening tool for sepsis without laboratory results in the emergency department: a machine learning study. *eClinicalMedicine*. 2025 Feb 1;80. doi:[10.1016/j.eclinm.2024.103048](https://doi.org/10.1016/j.eclinm.2024.103048) PubMed PMID: 39867965.
8. Liu Z, Shu W, Li T, Zhang X, Chong W. Interpretable machine learning for predicting sepsis risk in emergency triage patients. *Sci Rep*. 2025 Jan 6;15(1):887. doi:[10.1038/s41598-025-85121-z](https://doi.org/10.1038/s41598-025-85121-z)

General AI-assisted Triage Research Paper Summaries

Summary 9- Use of Artificial Intelligence in Triage in Hospital Emergency Departments: A Scoping Review

Tyler et al. (2024) researched whether machine learning models improve clinical outcomes and triage accuracy in emergency departments compared to traditional five-level triage systems that rely on nurses' subjective judgment. Using a scoping review based on Joanna Briggs Institute methodology, the researchers searched EMBASE, Ovid MEDLINE, and Web of Science for US-based, peer-reviewed primary research studies published between 2013 and 2023, screening 1,140 citations down to 29 included studies. Overall, the findings showed that machine learning models consistently outperformed conventional triage systems in areas such as predictive accuracy, disease identification, risk assessment, resource allocation and prediction of length of stay. However, the authors noted that the review was limited to studies published in the United States, which may reduce how well the findings apply to healthcare systems with different staffing structures, patient populations, and resource availability. This represents a direct gap for Mercer General Hospital, a Caribbean emergency department with a distinct patient mix and resource constraints not considered in the 29 reviewed studies.

Summary 10- AI-driven triage in emergency departments: A review of benefits, challenges, and future directions

Da'Costa et al. (2025) explored whether artificial intelligence–driven triage systems can improve emergency department patient prioritisation compared to traditional triage methods that rely on subjective clinical judgment, particularly during periods of overcrowding and mass-casualty events. To investigate this, the authors conducted a narrative review of peer-reviewed literature published between 2014 and 2024, using databases including PubMed, Scopus, IEEE Xplore, and Google Scholar. The findings suggested that machine learning triage systems can significantly improve patient prioritisation, reduce wait times, and optimise resource allocation within emergency departments. However, the author notes that the narrative review design limits quantitative assessment of evidence strength, and may exclude relevant older, non-English, and grey literature. The authors also noted possible publication bias and limited attention to region-specific healthcare differences, which may reduce the generalisability of findings to settings such as Caribbean emergency departments.

Summary 11 - TriageIntelli: AI-Assisted Multimodal Triage System for Health Centers

Araouchi and Adda (2024) investigated whether combining machine learning (ML) models could predict emergency department triage levels more accurately than a single model, with the aim of helping clinicians prioritise patients more effectively during periods of overcrowding. To test this, they developed and compared seven ML models (Support Vector Machine, Random Forest, Artificial Neural Network, Gradient Boosting Machine, Logistic Regression, XGBoost, and a stacking ensemble model that combined the three best-performing algorithms) using a Korean emergency department dataset labelled according to the Korean Triage and Acuity Scale (KTAS). The results showed that the combined model, TriageIntelli, achieved the highest predictive accuracy among all seven models, demonstrating that combining the outputs of multiple well-performing algorithms improved triage classification.

performance. However, the authors noted that the model was developed and tested only on Korean KTAS data and had not been validated in non-Korean populations. Differences in patient demographics, clinical practices, and documentation systems limit its generalisability to other healthcare settings.

References:

1. Tyler S, Olis M, Aust N, Patel L, Simon L, Triantafyllidis C, et al. Use of Artificial Intelligence in Triage in Hospital Emergency Departments: A Scoping Review. *Cureus*. 2024 May 8;16(5). doi:[10.7759/cureus.59906](https://doi.org/10.7759/cureus.59906)
2. Da'Costa A, Teke J, Origbo JE, Osonuga A, Egbon E, Olawade DB. AI-driven triage in emergency departments: A review of benefits, challenges, and future directions. *International Journal of Medical Informatics*. 2025 May 1;197:105838. doi:[10.1016/j.ijmedinf.2025.105838](https://doi.org/10.1016/j.ijmedinf.2025.105838)
3. Araouchi Z, Adda M. TriageIntelli: AI-Assisted Multimodal Triage System for Health Centers. *Procedia Computer Science*. 2024 Jan 1;15th International Conference on Emerging Ubiquitous Systems and Pervasive Networks / 14th International Conference on Current and Future Trends of Information and Communication Technologies in Healthcare EUSPN/ICTH 2024251:430–7. doi:[10.1016/j.procs.2024.11.130](https://doi.org/10.1016/j.procs.2024.11.130)